SHUMAIL FAROOQI

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EDUCATION

University of Toronto Graduating Dec 2027

BSc, Mathematical Physics Specialist & Actuarial Science Major GPA: 3.51/4.0

Relevant Coursework

- **Actuarial Science & Statistics:** Mathematics of Investment & Credit, Financial Principles for Actuarial Science I & II, Introductory Life Contingencies, Advanced Life Contingencies, Loss Models: From Data to Decisions, Statistical Methods for Actuarial Loss, Stochastic Methods for Finance, Probability and Statistics I & II
- **Mathematics:** Analysis I & II, Foundations of Real Analysis, Linear Algebra I & II, Advanced Ordinary Differential Equations, Partial Differential Equations, Complex Analysis I, Topology, Differential Geometry, Nonlinear Optimization, Applied Nonlinear Equations, Independent Reading in Mathematics
- **Physics:** Classical Mechanics, Advanced Classical Mechanics, Electricity & Magnetism, Electromagnetic Theory, Thermal Physics, Statistical Mechanics, Introduction to Quantum Mechanics, Quantum Mechanics I & II, Quantum Information, Mathematical Foundations of Quantum Mechanics and Quantum Information Theory, Practical Physics I & II, Advanced Physics Laboratory
- **Computer Science:** Introduction to Computer Science (object-oriented programming, data structures, recursion, algorithm analysis)

ACTUARIAL EXAMS

Exam FM (Financial Mathematics) – Passed December 2025

Exam P (Probability) – Passed July 2026

Exam MAS-I (Modern Actuarial Statistics I) – In Progress

TECHNICAL SKILLS

Programming: Python, SQL, VBA, MATLAB, HTML, CSS, JavaScript

Python Libraries: NumPy, SciPy, Pandas, Pygame

Actuarial & Analytics: Ratemaking, Loss Ratio Analysis, Frequency–Severity Analysis, Statistical Modeling, Curve Fitting, Uncertainty Analysis, Data Visualization

Tools: Excel (Pivot Tables, Dashboards), Access, QuickBooks, Jupyter Notebook, LaTeX, Git, GitHub, Microsoft Office (Word, Outlook, PowerPoint), Google Workspace

Professional: Oral and written communication, presentations, collaboration, interpersonal skills

EXPERIENCE

Summer Actuarial Student June 2026 – July 2026

CAS Student Central Summer Program

- Completed a competitive CAS program in P&C pricing, reserving, predictive modeling, and data visualization, working with actuarial mentors and a cohort of students
- Priced new homeowners discounts for water sensors, alarms, and roof types on a 200,000-policy, six-peril book as part of a 7-person team in the program's case competition
- Analyzed loss ratios by protective feature, finding policies with water sensors ran a 38.7% loss ratio versus 66.4% without

- Blended indicated relativities with market-share-weighted competitor discounts using square-root credibility (full credibility at 1,082 claims), and adjusted the solar-panel roof complement to reflect higher repair costs
- Rebased water, wind/hail, and theft base rates by 6.4%, 8.1%, and 24.2% to offset a $25.9M premium reduction from the new discounts, keeping total premium level with the prior year's $466.6M
- Presented the pricing recommendation to industry professionals

Finance Intern June 2025 – August 2025
Shepherd Village Scarborough, ON

- Reconciled monthly financial statements and identified reporting inconsistencies for management review
- Audited expense reports and vendor invoices to improve financial accuracy and detect irregularities
- Supported audit planning through financial data analysis and identification of operational risk areas
- Collaborated with finance and HR teams to streamline administrative workflows and improve data integrity

Accounting Student (Co-op) September 2020 – January 2022
Ahmadiyya Community Vaughan, ON

- Submitted invoices daily through the organization's financial platform to keep records up to date
- Cleaned financial data to ensure accuracy in records and reports
- Designed and delivered weekly financial reports with visual dashboards in Excel
- Organized and maintained physical and digital financial records for quick retrieval

CASE COMPETITIONS

AI Liability Risk & Insurance Strategy – Quantify Case Competition August 2026

- With a team of four, advised a mid-sized Canadian P&C insurer ($834M earned premium across five lines) on whether to enter the AI liability market
- Projected 2026 profit by line from five years of financial data, flagging Cyber Liability (28% of volume) as turning unprofitable as policy volume grew faster than premium
- Analyzed 135 AI incidents across six failure types for frequency, severity, and tail risk, finding autonomous system failures averaged $19.9M per loss, over 130× hallucinated output ($152K)
- Linked human oversight to 43% lower average losses ($2.40M vs. $4.23M), proposing it as an underwriting condition or pricing discount
- Recommended a standalone AI liability product, reinsured through Munich Re and excluding autonomous system failures at launch, while building proprietary claims data for future pricing

RESEARCH

K-Theory for C*-Algebras May 2024 – August 2024
University of Toronto & Fields Institute Toronto, ON

- Completed an independent reading course (MAT295) in operator algebras and K-theory, supervised by Prof. George A. Elliott
- Studied C*-algebras, which combine algebra and analysis to study spaces of operators, with ties to spectral theory and operator representations
- Constructed the K_0 group of a unital C*-algebra via the Grothendieck construction and presented it in an oral talk and written report at a student conference at the Fields Institute

PROJECTS

Energy Calibration for a Particle Detector February 2025

- Built pulse-amplitude estimators from simulated particle detector traces in Python (NumPy, SciPy, Pandas)
- Calibrated the estimators on a known 10 keV source, finding a calibration factor of 41.6 keV/mV, then used it to measure the energy spectrum of an unknown particle
- Fit energy spectra with statistical models using χ^2 goodness of fit and uncertainty analysis, and documented methods and findings in a formal scientific report

Damped Harmonic Motion Modeling April 2025

- Built a physical pendulum and collected oscillation time-series data through video processing in Python
- Fit period against string length with a reduced χ^2 of 1.89, confirming the predicted length dependence
- Estimated damping coefficients through nonlinear curve fitting, finding the decay constant's dependence on mass fit a logarithm better than the predicted linear model

Spectrometry Project February 2025

- Calibrated a spectrometer with mercury and measured the emission spectra of hydrogen and helium, co-authoring the report with a lab partner
- Measured hydrogen's Balmer line at 1.87 ± 0.05 eV against a theoretical 1.89 eV, verifying the Rydberg formula and quantum selection rules
- Identified an unknown gas as krypton from its three emission peaks

Portfolio Website 2026

- Designed and built a personal portfolio site from scratch in HTML, CSS, and JavaScript, hosted on GitHub Pages
- Documented projects and completed coursework with one shared stylesheet, self-hosted fonts, and a layout that works from phone to wide screen

TreeMap Visualizer

- Built an interactive treemap of directories and files in Python using Pygame and the os library, with trees as the core data structure
- Supported moving files and folders between directories directly from the treemap window

LEADERSHIP & VOLUNTEERING

Finance Executive August 2025 – May 2026
UofT Quantum Computing Club Toronto, ON

- Prepared funding applications using budget projections, cost estimates, and financial summaries for club initiatives
- Managed club reimbursements and monitored organizational spending to maintain budget oversight

Member *UofT Physics Union*

- Assisted with organizing student events and departmental activities for the physics community
- Supported event logistics including setup, coordination with organizers, and managing attendee participation

Volunteer *Humanity First Food Bank*

- Organized, packaged, and distributed food supplies to community members at local food bank distributions
- Worked with a team of volunteers to manage inventory and keep distribution running smoothly during high-volume events

ACTIVITIES & INTERESTS

Clubs: UofT Actuarial Science Club, UofT Quantum Computing Club, UofT Run Club, UofT Climbing Club
Interests: Rock climbing, running, video games